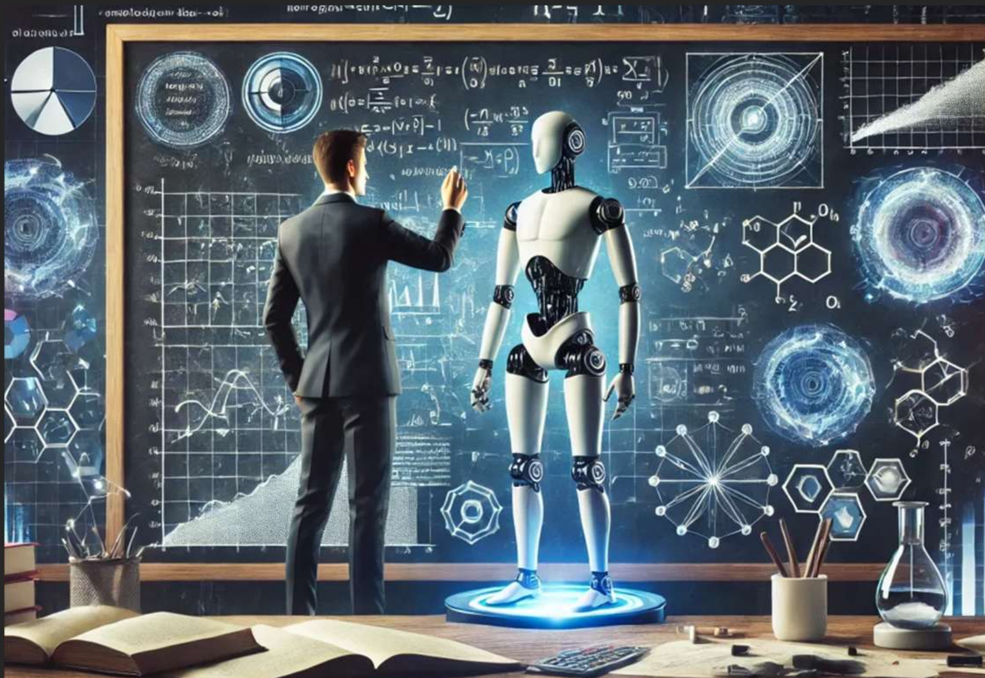
AI IN ACTION: MY APPLIED PROJECTS & INDUSTRY INSIGHTS

Aima Ayaz

INTRODUCTION

AI student with hands-on experience in machine learning, computer vision, and applied AI systems. Currently developing “Sick or Not” (teamed), an AI-powered temperature-based health screening system combining object detection, thermal analysis, and predictive modeling.



PROJECT OVERVIEW

Sick or Not — AI Temperature-Based Health Screening

- Addresses real-world needs highlighted during events like COVID-19.
- Uses AI to detect a person, extract temperature data, and classify whether the temperature is abnormal.
- Removes the need for manual temperature checks and supports early illness detection.

What I Learned

- Problem framing for real-world health applications
- Designing end-to-end AI systems (data → model → deployment)
- Working with video streams and multi-model pipelines.

TECHNICAL HIGHLIGHTS

Key Components

• YOLO Object Detection — Detects the person in the video stream.

• Temperature Overlay System — Maps thermal data onto the detected body region.

• Binary Classifier — Predicts “normal” vs “abnormal” temperature.

• Server + Camera Pipeline — Manages video input, model inference, and real-time output.

Tools & Concepts Mastered

• YOLO models • Binary classification • Data sourcing (Hugging Face, Kaggle) • Real-time inference pipelines • Precision, Recall, F1 Score evaluation

REAL-WORLD APPLICATIONS (HOW THIS PROJECT APPLIES TO INDUSTRY)

- **Healthcare:** Automated pre-screening in clinics, hospitals, and urgent care.
- **Public Safety:** Temperature checks in airports, schools, workplaces.
- **Smart Cities:** Integrating health monitoring into public infrastructure.
- **Automation:** Reduces manual labor and improves consistency in health screening.

INDUSTRY PROBLEMS IT SOLVES

- Removes human error in temperature screening
- Early detection of abnormal temperature
- Reducing human exposure to contagious individuals
- Faster, contactless screening during outbreaks

INFO

GitHub Portfolio: https://github.com/AimaAyaz/ITAI-2277-Capstone-Portfolio.git

Email: w216981450@student.hccs.edu

Skills: Machine Learning • Python • Data Analysis • AI Ethics • Case Study Research
Additional Work: NASA ML Project • Synthesia AI Abuse Case Study • Multi-industry AI Reports